AJINKYA AWARI

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PROFILE

MSc Artificial Intelligence for Biomedicine and Healthcare student at UCL, building reproducible biomedical AI and evaluation systems with explicit evidence and limitations, and reporting weak and null results openly.

EDUCATION

University College London (UCL), London, UK ***2026-2027***

MSc - Master of Science, Artificial Intelligence for Biomedicine and Healthcare

Savitribai Phule Pune University, Pune, IN | **CGPA: 8.25/10** ***2019-2023***

BE - Bachelor of Engineering, Computer Science

Leadership: Vice President - Technology Club

SKILLS

– **Languages:** Python, C, C++

– **ML / DL Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn

– **AI & Research Areas:** Graph Neural Networks, Protein Language Models, Federated Learning, Model Evaluation, Explainable AI, Reproducible Research

– **Data Science Tools and Technologies:** Hadoop, Apache Spark, SQL, BigQuery, Docker, Matplotlib

– **Web & Tools:** HTML, CSS, Bootstrap, Flask, Streamlit, Git, AWS, GCP

EXPERIENCE

Byju's Think & Learn Pvt. Ltd, India ***July 2023 – July 2024***

Cloud Engineer

– Implemented a predictive analytics pipeline on cloud platforms using Python, enabling sales teams to prioritise high-potential leads, improving lead-to-customer conversion by 12% and driving an 8% increase in quarterly revenue.

– Designed and optimised a scalable cloud infrastructure capable of handling 300,000+ concurrent users by improving backend architecture, database queries, and load balancing, reducing latency by 15% and ensuring 99.7% uptime during peak enrolments.

– Optimised cloud-based lead management automation by integrating LeadSquared, reducing manual effort by 4+ hours per week and boosting lead conversion rates by 10%, contributing to $300K+ in additional revenue. Utilised cloud monitoring and performance metrics to refine iterations of infrastructure and automation.

The Leadership 30 ***July 2024 – August 2026***

Senior Data Analyst

– Disaster Resilience & AI-Driven Risk Mitigation: Built an AI-powered flood forecasting system in collaboration with local agencies, enhancing emergency response times by 12% and safeguarding many individuals in 6 flood-prone districts across Maharashtra.

– Transforming Education in Maharashtra (SDG 4 - Quality Education): Led a data-driven initiative with the Maharashtra government on a three-month project to assess learning gaps in public schools. Developed targeted intervention strategies that boosted student engagement by 15% and improved literacy outcomes for 2,000+ students in rural districts.

– Advancing Clean Energy Access (SDG 7 - Affordable and Clean Energy): Designed a cost-benefit analysis model for renewable energy projects in rural Maharashtra, facilitating policy recommendations for solar micro grids. The initiative led to a 10% increase in clean energy adoption, benefiting 800+ households and reducing annual carbon emissions by 1,500 tons.

PROJECTS

AIXI-Complexity Benchmark

– Built a local CPU benchmark comparing five decision-making agents (a random policy, tabular Q-learning, an AIXI-inspired planner, a small local language model and an exact dynamic-programming oracle) on 20 frozen finite-state tasks, with 10 replicates per task and fixed seeds.

– Evaluated agents with a normalised oracle-gap metric, a 10,000-resample paired bootstrap per task and hierarchical bootstrap intervals for the suite, supported by 81 tests, a clean lint run and passing continuous integration.

– Applied Holm correction across difficulty proxies and reported the null result openly: no proxy association remained significant (minimum adjusted $p = 0.144$), and the language-model arm was valid on only 10 of the 20 tasks.

GraphGPS-Upgrade

– Built a fixture-first graph-level benchmark (graph batching, graph-local positional features and uniform PyTorch Geometric baselines) to compare GraphGPS, GCN and GIN fairly on the official ogbg-molhiv scaffold split.

– Ran a full-budget ablation on a Tesla T4 under one frozen recipe (30 epochs maximum, patience 5, three seeds), giving test ROC-AUC of 0.748 for GraphGPS against 0.698 for GIN and 0.693 for GCN.

– Limited the claim to this frozen recipe and recorded the blocked and one-epoch smoke runs separately, rather than presenting them as benchmark results.

ESM2 Protein Fitness Benchmark

– Built a reproducible, leakage-aware benchmark comparing frozen ESM1v and ESM2 representations against simple baselines on ProteinGym single-substitution assays, grouping splits by UniProt ID and sequence-similarity clusters with SHA-256-hashed split manifests.

– Evaluated 40 assays on CPU: ESM1v (five checkpoints, masked-marginal scoring) reached a macro Spearman of 0.475, against 0.266 for frozen ESM2-650M embeddings with ridge regression.

– Covered the pipeline with 112 tests and a privacy-gated public results boundary; grouped bootstrap confidence intervals remain pending GPU access.

PUBLICATIONS

V. Paithankar, A. Awari (second author), A. Raskar, S. Patil, N. Jamdar. *Plant Disease Detection using Machine Learning.* IJARSCT, Vol. 3, Issue 2 (doi:10.48175/IJARSCT-9156) and Issue 4 (doi:10.48175/IJARSCT-9297), April 2023.